

KAZANARYK, Turkmenistan

WMO: 388060

Lat: 38.467N

Lon: 64.367E

Elev: 212

StdP: 98.80

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-5.3	-12.1	1.4	-3.0	-9.6	1.7	-0.3	8.1	13.8	6.7	11.9	1.5	0	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	40.5	21.3	39.1	20.7	37.9	20.2	23.4	36.7	22.2	35.7	21.3	34.7	4.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.1	30.7	17.6	13.0	28.7	16.5	12.1	27.5	70.9	36.7	66.4	35.7	62.9	34.9	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.3	5.8	-11.8	43.3	4.3	1.4	-14.9	44.3	-17.4	45.2	-19.8	45.9	-22.9	47.0		
			DB													
			WB	-12.2	26.3	4.4	2.1	-15.3	27.8	-17.9	29.0	-20.3	30.1	-23.5	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	4.8	6.3	12.1	18.2	24.0	28.2	29.7	27.7	22.3	16.1	9.5	5.0	(6)
(7)		DBStd	9.93	4.63	5.41	5.01	4.81	3.33	2.68	2.59	2.92	3.40	4.51	5.24	4.84	(7)
(8)		HDD10.0	578	172	123	36	3	0	0	0	0	7	70	168		(8)
(9)		HDD18.3	1807	421	337	200	61	3	0	0	7	96	268	414		(9)
(10)		CDD10.0	3150	9	19	102	250	434	545	611	549	368	197	55	12	(10)
(11)		CDD18.3	1337	0	0	7	58	178	295	352	291	125	27	3	0	(11)
(12)		CDH23.3	19464	0	2	79	653	2317	4469	5553	4247	1688	418	37	1	(12)
(13)		CDH26.7	11431	0	0	14	239	1169	2777	3609	2653	836	127	6	0	(13)
(14)	Wind	WSAvg	2.3	2.2	2.4	2.4	2.3	2.4	2.7	3.0	2.6	2.0	1.9	1.9	2.0	(14)
(15)	Precipitation	PrecAvg	169	26	32	32	23	11	1	0	1	4	18	23		(15)
(16)		PrecMax	298	72	97	77	108	48	8	10	4	4	28	49	76	(16)
(17)		PrecMin	100	5	2	5	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	55	15	25	15	22	11	2	2	1	1	6	12	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.5	22.1	28.9	34.9	38.2	41.1	43.0	41.4	37.7	32.8	27.5	20.5	(19)
(20)		MCWB	10.1	11.4	15.4	18.6	19.8	21.8	23.1	21.1	19.7	17.3	14.7	10.4		(20)
(21)		2%	DB	16.5	19.2	25.6	31.9	36.3	39.4	41.1	39.6	35.6	29.8	23.6	17.5	(21)
(22)		MCWB	8.9	10.7	14.2	17.4	19.1	21.0	21.5	20.7	19.5	16.2	13.5	9.2		(22)
(23)		5%	DB	14.3	16.8	23.1	29.5	34.5	38.2	39.5	38.2	33.7	27.7	20.6	15.0	(23)
(24)		MCWB	8.1	9.7	13.3	16.6	18.6	20.4	20.8	20.3	18.6	15.3	12.3	8.5		(24)
(25)		10%	DB	11.8	14.5	20.5	27.3	32.7	36.8	38.1	36.7	31.5	25.5	18.2	12.5	(25)
(26)		MCWB	7.1	8.8	12.2	15.8	18.1	19.8	20.4	20.0	17.8	14.7	11.0	7.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	13.2	16.6	20.8	21.8	25.6	25.8	24.7	21.6	18.6	15.8	11.7	(27)
(28)		MDCB	16.9	19.5	25.7	30.7	33.5	37.7	39.2	37.6	34.7	29.7	24.2	17.2		(28)
(29)		2%	WB	9.7	11.7	15.2	19.0	20.6	23.1	23.4	22.7	20.3	17.2	14.2	10.1	(29)
(30)		MDCB	15.1	17.1	23.1	28.1	32.6	36.1	37.0	36.0	33.2	27.4	21.7	16.0		(30)
(31)		5%	WB	8.6	10.4	14.0	17.7	19.7	21.7	22.3	21.6	19.2	16.1	12.8	8.9	(31)
(32)		MDCB	13.4	15.6	21.2	26.4	31.5	35.3	36.1	34.9	31.9	26.1	19.3	14.2		(32)
(33)		10%	WB	7.4	9.2	12.9	16.7	18.9	20.7	21.2	20.6	18.2	15.0	11.6	7.7	(33)
(34)		MDCB	11.5	14.2	19.5	25.4	30.3	34.1	35.1	34.1	30.2	24.4	17.3	12.1		(34)
(35)	Mean Daily Temperature Range	MDBR	9.1	10.4	11.3	12.8	14.7	15.6	15.2	16.5	17.4	15.8	12.0	9.5		(35)
(36)		5% DB	MCDDBR	12.3	13.4	14.6	16.2	17.3	17.5	17.0	18.0	19.5	18.0	15.0	12.7	(36)
(37)		MCWBR	6.6	6.8	6.1	6.1	6.1	6.1	5.6	6.1	7.6	7.7	7.1	6.8		(37)
(38)		5% WB	MCDDBR	11.0	11.4	12.7	13.8	15.1	16.3	16.0	17.6	18.7	16.7	13.5	11.4	(38)
(39)		MCWBR	6.4	6.2	5.9	5.8	5.9	6.8	6.6	7.0	7.9	8.0	6.8	6.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.390	0.432	0.501	0.525	0.523	0.525	0.499	0.461	0.440	0.477	0.446	0.393		(40)
(41)		taud	2.092	1.988	1.851	1.830	1.859	1.848	1.909	1.994	2.011	1.894	1.962	2.069		(41)
(42)		Ebn at Noon	747	765	750	763	775	771	788	809	799	704	669	705		(42)
(43)		Edh at Noon	118	149	189	204	202	205	191	171	158	161	133	113		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.13	3.06	4.21	5.62	6.98	7.92	7.77	7.14	5.90	4.11	2.63	1.93		(44)
(45)		RadStd	0.21	0.31	0.31	0.45	0.29	0.21	0.14	0.15	0.15	0.24	0.24	0.21		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.39	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.58	-0.80	N/A	N/A	N/A	+52

Nomenclature: See separate page